

Rahul Rawat

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Airbus Defence and Space

Hiring Team

Application: Working Student, Development of AI Models for Spacecraft FDIR

Dear Hiring Team,

I am applying for the Working Student position developing AI models for Spacecraft FDIR with the Digital Payload Processors Design Team at Airbus Defence and Space in Taufkirchen. The brief on exploring AI architectures for input driven results, running trade off analysis on key performance indices, and validating AI models to ensure reliability and performance is exactly the shape of work I have been shipping.

At eRay GmbH I built an end to end recursive time series pipeline forecasting four water quality indicators for a German lake, benchmarking six models head to head, running quantile regression for asymmetric 80 percent prediction intervals, and enforcing strict anti leakage rules across the pipeline. In CreditIQ I lifted the Disparate Impact ratio from 0.79 to 0.88 and cut the false negative rate from 44 to 16.7 percent while holding accuracy, backed by unit tests at 100 percent branch coverage. That is directly the trade off analysis and validation discipline autonomous FDIR needs.

My Hybrid RAG Orchestrator is a working AI system with a custom decision making router over Llama 3.1 8b via Groq and LangChain, showing I can design and integrate AI components into a larger inference pipeline. In my Real Time Flight Tracking Data Pipeline I processed over 128 thousand records with PySpark on Google Cloud, enriching flight positions against airport, aircraft, and weather data every 30 seconds, which gives me a working feel for aerospace grade signal fusion.

I am fluent in English, comfortable in Python and neural network modelling frameworks, and hold the NVIDIA Building LLM Applications With Prompt Engineering certificate. I would be glad to start on 1 October 2026 for a minimum of six months at 18 hours per week from the Taufkirchen site.

Kind regards,

Rahul Rawat